

FORM TP 2013236



TEST CODE **02234020**

MAY/JUNE 2013

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

PURE MATHEMATICS

UNIT 2 – Paper 02

ANALYSIS, MATRICES AND COMPLEX NUMBERS

2 hours 30 minutes

29 MAY 2013 (p.m.)

This examination paper consists of **THREE** sections: Module 1, Module 2 and Module 3.

Each section consists of 2 questions.

The maximum mark for each Module is 50.

The maximum mark for this examination is 150.

This examination consists of 6 printed pages.

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. **DO NOT** open this examination paper until instructed to do so.
2. Answer **ALL** questions from the **THREE** sections.
3. Write your solutions, with full working, in the answer booklet provided.
4. Unless otherwise stated in the question, any numerical answer that is not exact **MUST** be written correct to three significant figures.

Examination Materials Permitted

Graph paper (provided)

Mathematical formulae and tables (provided) – **Revised 2012**

Mathematical instruments

Silent, non-programmable, electronic calculator

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02234020/CAPE 2013



SECTION A (Module 1)

Answer BOTH questions.

1. (a) Calculate the gradient of the curve $\ln(x^2y) - \sin y = 3x - 2y$ at the point $(1, 0)$. **[5 marks]**

- (b) Let $f(x, y, z) = 3yz^2 - e^{4x} \cos 4z - 3y^2 - 4 = 0$.

Given that $\frac{\partial z}{\partial y} = -\frac{\partial f / \partial y}{\partial f / \partial z}$, determine $\frac{\partial z}{\partial y}$ in terms of x, y and z . **[5 marks]**

- (c) Use de Moivre's theorem to prove that

$$\cos 5\theta = 16 \cos^5 \theta - 20 \cos^3 \theta + 5 \cos \theta. \quad \text{[6 marks]}$$

- (d) (i) Write the complex number $z = (-1 + i)^7$ in the form $re^{i\theta}$, where $r = |z|$ and $\theta = \arg z$. **[3 marks]**

- (ii) Hence, prove that $(-1 + i)^7 = -8(1 + i)$. **[6 marks]**

Total 25 marks

2. (a) (i) Determine $\int \sin x \cos 2x \, dx$. **[5 marks]**

- (ii) Hence, calculate $\int_0^{\frac{\pi}{2}} \sin x \cos 2x \, dx$. **[2 marks]**

- (b) Let $f(x) = x|x| = \begin{cases} x^2 & ; x \geq 0 \\ -x^2 & ; x < 0 \end{cases}$.

Use the trapezium rule with four intervals to calculate the area between $f(x)$ and the x -axis for the domain $-0.75 \leq x \leq 2.25$. **[5 marks]**

- (c) (i) Show that $\frac{2x^2 + 4}{(x^2 + 4)^2} = \frac{2}{x^2 + 4} - \frac{4}{(x^2 + 4)^2}$. **[6 marks]**

- (ii) Hence, find $\int \frac{2x^2 + 4}{(x^2 + 4)^2} \, dx$. Use the substitution $x = 2 \tan \theta$. **[7 marks]**

Total 25 marks

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SECTION B (Module 2)

Answer BOTH questions.

3. (a) The sequence $\{a_n\}$ is defined by $a_1 = 1$, $a_{n+1} = 4 + 2\sqrt[3]{a_n}$.

Use mathematical induction to prove that $1 \leq a_n \leq 8$ for all n in the set of positive integers. **[6 marks]**

- (b) Let $k > 0$ and let $f(k) = \frac{1}{k^2}$.

(i) Show that

a) $f(k) - f(k+1) = \frac{2k+1}{k^2(k+1)^2}$. **[3 marks]**

b) $\sum_{k=1}^n \left(\frac{1}{k^2} - \frac{1}{(k+1)^2} \right) = 1 - \frac{1}{(n+1)^2}$. **[5 marks]**

(iii) Hence, or otherwise, prove that

$$\sum_{k=1}^{\infty} \frac{2k+1}{k^2(k+1)^2} = 1. \quad \textbf{[3 marks]}$$

- (c) (i) Obtain the first four non-zero terms of the Taylor Series expansion of $\cos x$ in ascending powers of $(x - \frac{\pi}{4})$. **[5 marks]**

- (ii) Hence, calculate an approximation to $\cos(\frac{\pi}{16})$. **[3 marks]**

Total 25 marks

4. (a) (i) Obtain the binomial expansion of

$$\sqrt[4]{(1+x)} + \sqrt[4]{(1-x)}$$

up to the term containing x^2 .

[4 marks]

- (ii) Hence, by letting $x = \frac{1}{16}$, compute an approximation of $\sqrt[4]{17} + \sqrt[4]{15}$ to four decimal places.

[4 marks]

- (b) Show that the coefficient of the x^5 term of the product $(x+2)^5(x-2)^4$ is 96. [7 marks]

- (c) (i) Use the Intermediate Value Theorem to prove that $x^3 = 25$ has at least one root in the interval $[2, 3]$. [3 marks]

- (ii) The table below shows the results of the first four iterations in the estimation of the root of $f(x) = x^3 - 25 = 0$ using interval bisection.

The procedure used $a = 2$ and $b = 3$ as the starting points and p_n is the estimate of the root for the n^{th} iteration.

n	a_n	b_n	p_n	$f(p_n)$
1	2	3	2.5	-9.375
2	2.5	3	2.75	-4.2031
3	2.75	3	2.875	-1.2363
4	2.875	3	2.9375	0.3474
5				
6				
.....				
.....				

Complete the table to obtain an approximation of the root of the equation $x^3 = 25$ correct to 2 decimal places.

[7 marks]

Total 25 marks

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SECTION C (Module 3)

Answer BOTH questions.

5. (a) Three letters from the word BRIDGE are selected one after the other without replacement. When a letter is selected, it is classified as either a vowel (V) or a consonant (C).

Use a tree diagram to show the possible outcomes (vowel or consonant) of the THREE selections. Show all probabilities on the diagram. **[7 marks]**

- (b) (i) The augmented matrix for a system of three linear equations with variables x , y and z respectively is

$$A = \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ -5 & 1 & 1 & 2 \\ 1 & -5 & 3 & 3 \end{array} \right)$$

By reducing the augmented matrix to echelon form, determine whether or not the system of linear equations is consistent. **[5 marks]**

- (ii) The augmented matrix for another system is formed by replacing the THIRD row of A in (i) above with $(1 \ -5 \ 5 \ | \ 3)$.

Determine whether the solution of the new system is unique. Give a reason for your answer. **[5 marks]**

- (c) A country, X , has three airports (A , B , C). The percentage of travellers that use each of the airports is 45%, 30% and 25% respectively. Given that a traveller has a weapon in his/her possession, the probability of being caught is, 0.7, 0.9 and 0.85 for airports A , B , and C respectively.

Let the event that:

- the traveller is caught be denoted by D , and
- the event that airport A , B , or C is used be denoted by A , B , and C respectively.

- (i) What is the probability that a traveller using an airport in Country X is caught with a weapon? **[5 marks]**
- (ii) On a particular day, a traveller was caught carrying a weapon at an airport in Country X . What is the probability that the traveller used airport C ? **[3 marks]**

Total 25 marks

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6. (a) (i) Obtain the general solution of the differential equation

$$\cos x \frac{dy}{dx} + y \sin x = 2x \cos^2 x. \quad [7 \text{ marks}]$$

- (ii) Hence, given that $y = \frac{15\sqrt{2}\pi^2}{32}$, when $x = \frac{\pi}{4}$, determine the constant of the integration. [5 marks]

- (b) The general solution of the differential equation

$$y'' + 2y' + 5y = 4 \sin 2t$$

is $y = CF + PI$, where CF is the complementary function and PI is a particular integral.

- (i) a) Calculate the roots of

$$\lambda^2 + 2\lambda + 5 = 0, \text{ the auxiliary equation.} \quad [2 \text{ marks}]$$

- b) Hence, obtain the complementary function (CF), the general solution of

$$y'' + 2y' + 5y = 0. \quad [3 \text{ marks}]$$

- (ii) Given that the form of the particular integral (PI) is

$$u_p(t) = A \cos 2t + B \sin 2t,$$

$$\text{Show that } A = -\frac{16}{17} \text{ and } B = \frac{4}{17}. \quad [3 \text{ marks}]$$

- (iii) Given that $y(0) = 0.04$ and $y'(0) = 0$, obtain the general solution of the differential equation. [5 marks]

Total 25 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.